

Optimizing Product Assortment and Pricing Strategy for



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How BigBasket Works?

01



SHOP ONLINE

Customers order groceries via BigBasket app or website across multiple categories

WIDE ASSORTMENT

Offers products from top brands and local vendors, covering all price segments



02

03



FAST DELIVERY

Uses local warehouses and delivery partners to ensure timely doorstep delivery

Project Overview

Conduct exploratory analysis of BigBasket's product catalog to uncover insights related to assortment, pricing, brand positioning and market opportunities.



Data Overview

The dataset used in this analysis comprises 28,000+ products listed on BigBasket's platform. Each record represents a unique product offering and includes a variety of attributes related to product identity, categorization, pricing, and brand information.

✨ Key Features:

1 **23,510** unique products 

2 **11** categories 

3 **90** subcategories 

4 **2,312** unique brands 

~31%

Missing values for the “ratings” column which might indicate missing values or new products which isn’t rated yet

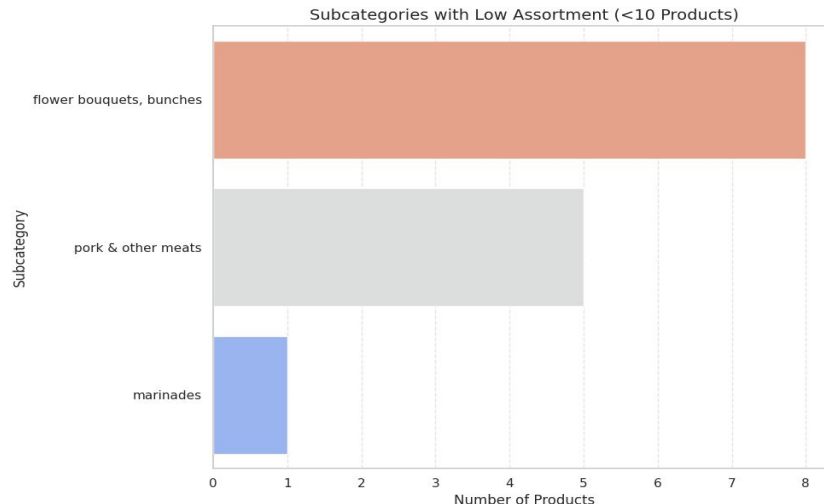
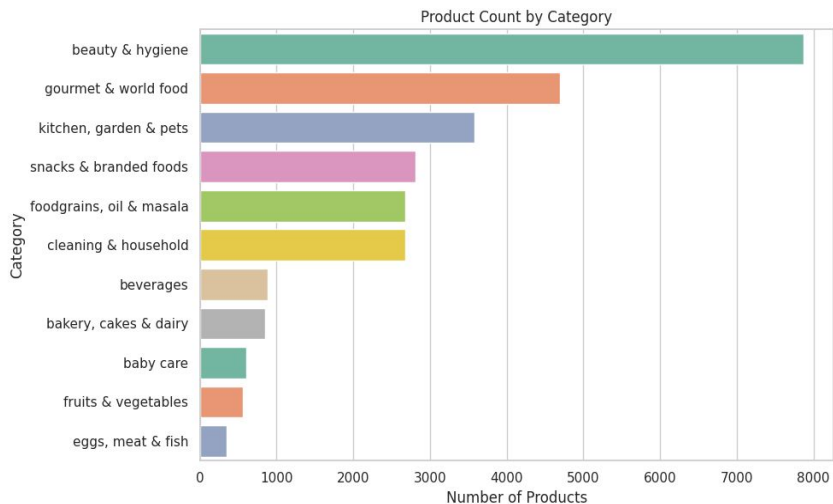
~6%

Missing values for the “description” column which was imputed

Data Cleaning Operations

1.  Handling Missing Values
2.  Dropping Redundant Columns
3.  Standardizing Column Names
4.  Data Type Optimization
5.  Feature Engineering
6.  Data Validation
7.  Categorical Normalization
8.  Outlier Handling

Product Assortment Optimization



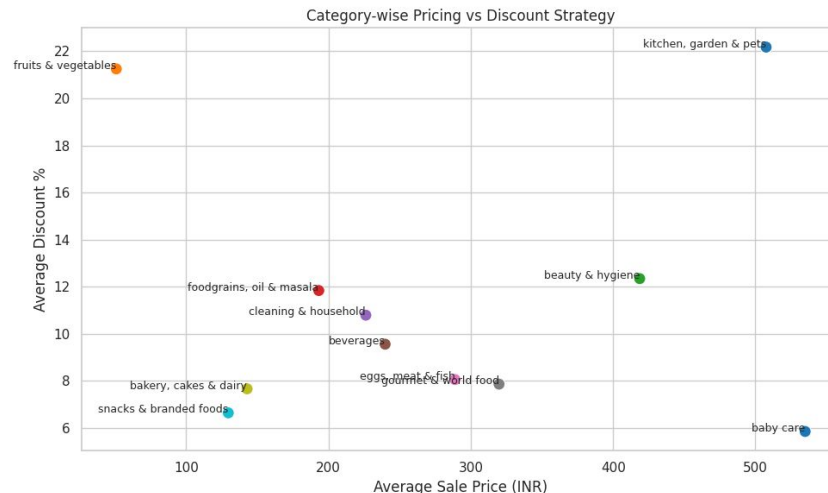
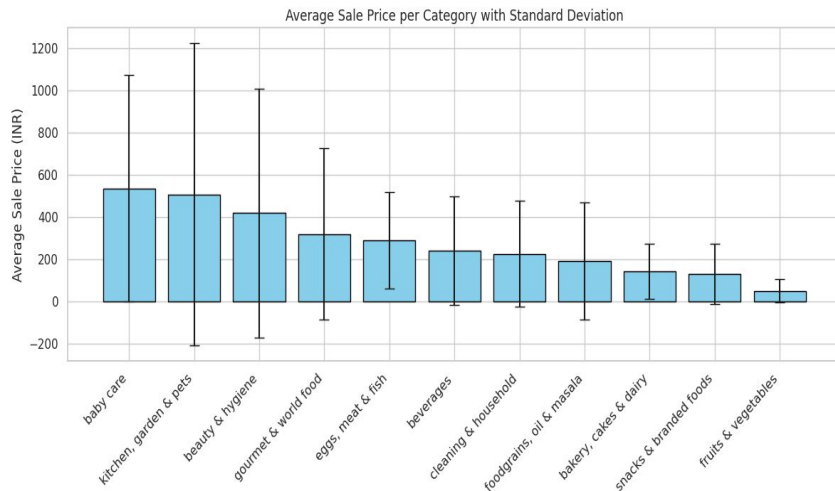
Key Insights:

- 📦 23,510 unique products across multiple categories.
- 💧 Categories with high SKU density: Beauty & Hygiene, Gourmet Foods.
- 🍖 Low-assortment subcategories: Marinades, Mutton & Lamb.

Actions:

- ✂️ Perform SKU rationalization.
- ✚️ Expand assortment in underserved segments.

Pricing Strategy Analysis



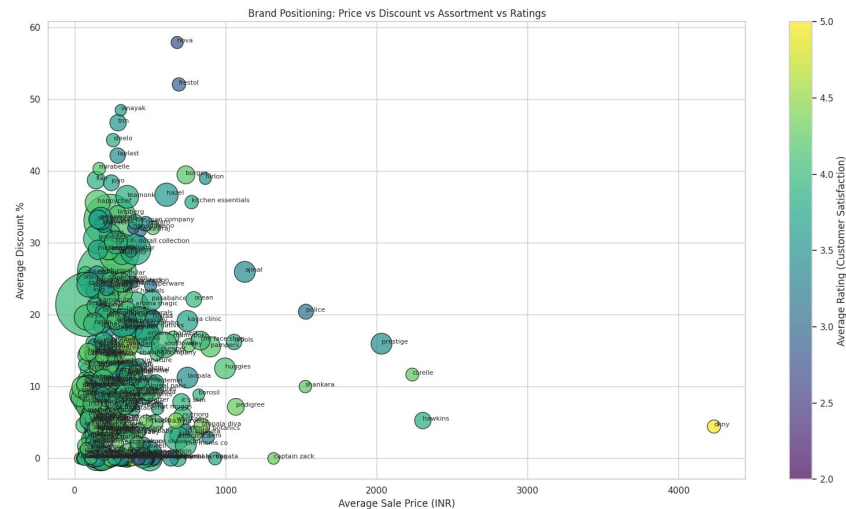
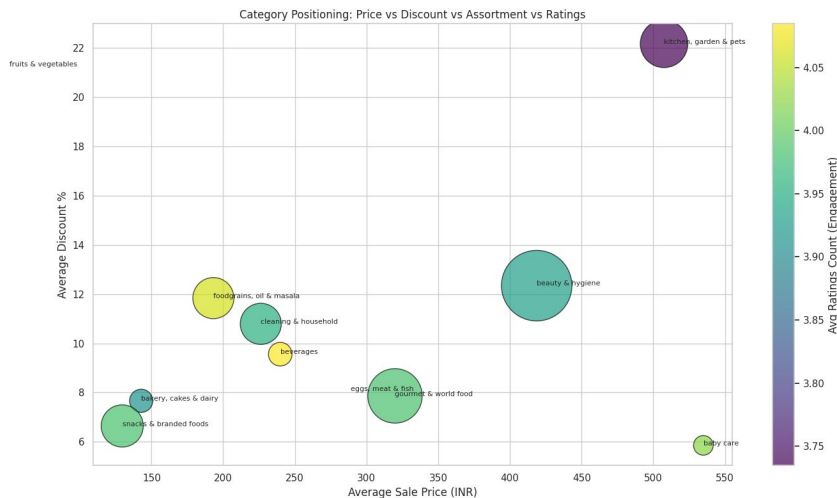
Key Insights:

- 🧠 Categories like Baby Care and Kitchen, Garden & Pets have premium pricing.
- 🌱 Essential categories like Food Grains have low pricing.
- 📊 Pricing vs Discount Strategy Breakdown.

Actions:

- 🔄 Reassess discount strategies for premium categories.
- 📊 Optimize discounts for essential, high-volume categories.

Brand & Category Positioning



Category-Level Positioning:

-  Premium categories: Baby Care, Gourmet Foods.
-  Mass-market categories: Food Grains, Snacks.

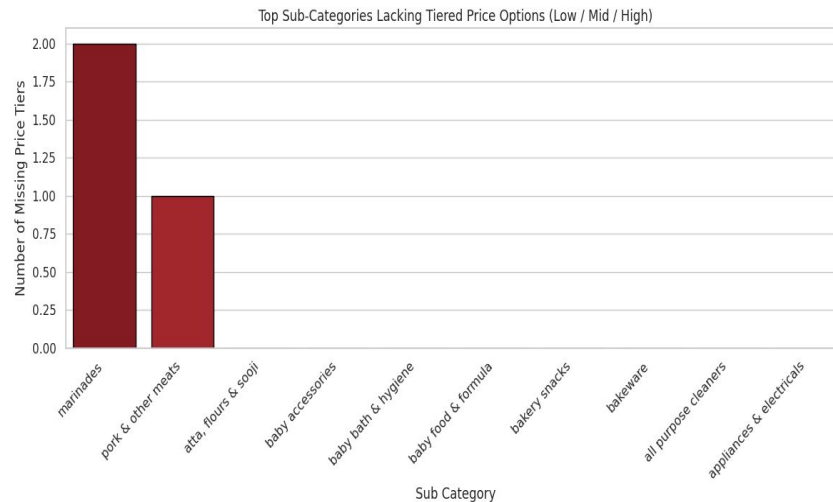
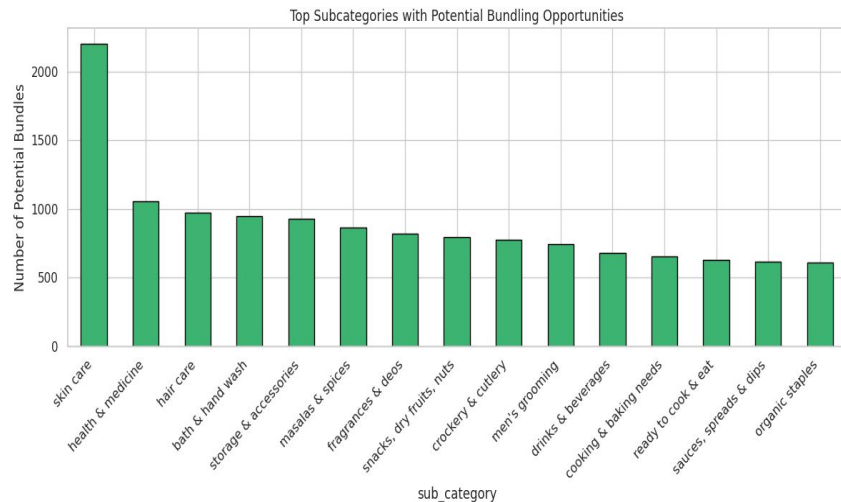
Actions:

-  Promote high-performing mid-tier brands.
-  Address low-rated, discount-heavy brands.

Brand-Level Positioning:

-  Premium brands: Hawkins, Corelle.
-  Discount-heavy brands: Nova, Bristol.
-  Mid-tier, high-performing brands: Amul, Tata.

Opportunity & Gap Identification



Key Findings:

-  High potential for bundling in Skin Care, Hair Care, and Health & Medicine.
-  Tier gaps in subcategories like Marinades and Pork.

Actions:

-  Expand brand presence in underserved subcategories.
-  Introduce tiered pricing options for specific segments.

Strategic Recommendations:

01



Rationalize Overcrowded SKUs in categories like Beauty & Hygiene.

02



Expand Assortment in niche or underdeveloped categories.

03



Optimize Discounting based on price elasticity and category positioning.

04



Create Bundles for value perception and increased average order value (AOV).

05



Boost Mid-Tier Brands with strong customer ratings.

06



Fill Tier Gaps to serve broader customer segments across subcategories.

Conclusion

Through a detailed analysis of BigBasket's product assortment, pricing, brand, and category strategies, we uncovered key areas of optimization and opportunity:

✅ **Refine Assortment** – Reduce SKU clutter in overcrowded categories and expand in niche, underserved segments.

💰 **Smart Pricing** – Align discounting strategies with product elasticity and category positioning to boost profitability.

🛍️ **Boost Brand Strategy** – Elevate high-potential mid-tier brands and streamline underperformers for better market impact.

📦 **Bundle & Tier Smartly** – Create value through intelligent bundling and ensure complete price coverage across subcategories.